**Course Project – Spring 2026**
**Object Oriented – Coffee Shop**:

For this project, you're going to develop an app, **using Java,** to help coffee shop owners offer their menu online. Create a new Project in IntelliJ named "Project2140".

The Main program (aka driver program):

**a)** Name the main program as following: **<your_name>**_Driver.java. Place a header with your name, date, and objective (a summary of what the program does) – **10 pt**

b) Main program will have two parts
   a. In the first part, the program prompts the vendor to enter the variety of coffee and flavors available for that day and their prices. – **10 pts**
   b. In the second part, each customer is asked to enter their name and to select the type of coffee they want and the flavor, then show the customer's balance. – **10 pts**

First Part example:

- Hello shop owner!
- Enter a coffee type (type "end" to stop):
- Cappuccino
- Enter a coffee type (type "end" to stop):
- Latte
- Enter a coffee type (type "end" to stop):
- Americano
- Enter a coffee type (type "end" to stop):
- Frappuccino
- Enter a coffee type (type "end" to stop):
- end
- Enter price for Cappuccino:
- 5.50
- Enter price for Latte:
- 4.50
- Enter price for Americano:
- 3.50
- Enter price for Frappuccino:
- 6.50
- Enter a flavor (type "end" to stop):
- Cream
- Enter price for Cream:
- 0.50

- Enter a flavor (type “end” to stop):
- Vanilla
- Enter price for Vanilla:
- 1.50
- Enter a flavor (type “end” to stop):
- Caramel
- Enter price for Caramel:
- 2.50
- Enter a flavor (type “end” to stop):
- end

Second Part example:

- Welcome to <your_name> Coffee Shop:
- Enter your name:
- Joe
- Choose a coffee type (or type 0 to end):
    - 1. Cappuccino
    - 2. Latte
    - 3. Americano
    - 4. Frappuccino
    - 0. End
- 1
- Choose a coffee type (or type 0 to end):
    - 1. Cappuccino
    - 2. Latte
    - 3. Americano
    - 4. Frappuccino
    - 0. End
- 0
- **Choose a flavor (or type 0 to end):**
    - 1. Cream
    - 2. Vanilla
    - 3. Caramel
    - 0. End
- 2
- **Choose a flavor (or type 0 to end):**
    - 1. Cream
    - 2. Vanilla
    - 3. Caramel
    - 0. End
- 0
- Your balance is: 7.00
- Please, come back and see us

Test part 1 by entering the values above and inserting them into the respective arrays created. Then, print each array.

Next, you are going to write the classes that you need for this project. Place each class in a separate IntelliJ src file (same project).

**Classes and Methods:**

1. For each class created, write a constructor with parameters and a no-parameter constructor; **15 pts**
2. Write getters/setters **as needed** – **10 pts**
3. Create at minimum **classes Coffee, Flavor, Customer**, and **ShoppingTray** and the respective objects – **20 pts**
4. **The menu shown in part 2 will be presented to the customer as many times as needed, as long as the customer selects a valid item in the menu. This is true for both Coffee and Flavor. For each item selected, an object is created and entered in either the Coffee arrayList or the Flavor arrayList, accordingly**
5. **ShoppingTray should contain a Coffee arrayList and a Flavor arrayList (besides Customer) as instance variables. It does not need a "quantity" variable. Instead, the arrayLists will provide how many coffee and how many flavors are being purchased. Also, it should contain a method to calculate the customer balance and to display the balance – 10 pts**
6. repeat the menu for the next customer – **10 pts**
7. The program should stop by typing "**end**" for customer's name; - **5 pts**
8. Test your classes before proceeding to the driver program.
9. Optional/extra: use input validation to ensure that the name and numbers entered by the customer are correct and within the range allowed by the program – **20 pts**

## Important:

1. You're going to partner with a classmate for this project. **Both team members must submit any milestone and the final program to D2L for the record**.
2. It's okay to discuss the project with other classmates, but it is NOT okay to send your entire program (or parts of it) to someone other than your partner;
3. **Two programs (from different teams) with exactly the same statements will be considered "plagiarism" and will be penalized;**
4. Please, follow the specifications closely to avoid deductions in your score
5. If a specification is not clear, ask me for clarification during class or email
6. **Use the Java statements, methods and style covered in this course; no "advanced" features or features not discussed in class (such as try/catch, recursion, etc) will be accepted**
7. **I will select some projects and have the authors explain their code to me in details.**